



Green University of Bangladesh

*Department of Computer Science and Engineering (CSE)
Semester: (Fall, Year: 2025), B.Sc. in CSE (Day)*

Snake Run

PROJECT PROPOSAL

*Course Title: Object Oriented Programming Lab
Course Code: CSE-202
Section: 242-D3*

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<u>Lab Project Status</u>	
Marks:	Signature:
Comments:	Date:

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Chapter 1

Introduction

1.1 Overview

This project is a complete recreation and enhancement of the classic Snake Game using Java Swing. The goal was to build a smooth, interactive, and visually appealing game with multiple modes, special items, AI-controlled movement, a settings panel, and persistent high-score saving. The game supports a graphical grid, animated effects, pop-up texts, shrinking food, bombs, and an AI mode where the snake moves automatically.

1.2 Motivation

I chose this project because game development helps strengthen core programming skills such as object-oriented design, event handling, and timer-based animation. Snake is simple yet complex enough to involve multiple engineering ideas—collision detection, pathfinding logic, dynamic rendering, and UI interaction. Another motivation was to create a polished game that others can actually play and enjoy.

1.3 Problem Definition

1.3.1 Problem Statement

The main problem addressed in this project was building a fully functional Snake game with modern enhancements. This includes designing smooth movement, preventing bugs in direction changes, adding special items with timers, implementing AI logic, and ensuring a clean UI with menus and settings. The challenge was to integrate everything seamlessly in Java Swing.

1.3.2 Complex Engineering Problem

Below is the required table based on the project characteristics.

Name of the Attributes	Explain how to address
P1: Depth of knowledge required	Required strong understanding of Java Swing, event handling, game loops, and rendering.
P2: Range of conflicting requirements	Balancing gameplay difficulty, speed, item lifetime, and AI decisions created conflicting constraints.
P3: Depth of analysis required	AI food-seeking and collision logic required careful analysis to avoid infinite loops or self-crashes.
P4: Familiarity of issues	Included debugging real-time movement, timer delays, UI flickering, and file saving errors.

Table 1.1: Placeholder caption for some Attributes and their explanations

1.4 Objectives

The design goals for this project are as follows:

- To build a fully playable Snake game using Java Swing.
- To implement a clean menu, settings screen, and score panel.
- To add special items (shrink food and bomb) with countdown timers.
- To create an AI mode where the snake intelligently moves toward food.
- To ensure smooth graphics with no flickering.
- To store high scores using file handling.

1.5 Application

A project like this reflects fundamental concepts of game programming. It can be used:

- As a learning resource for beginners in Java GUI development.
- As a base to build more advanced AI pathfinding (like A* or BFS).
- For educational demonstrations in real-time graphics and event loops.
- As a fun desktop game for entertainment.